

their ability to make informed decisions regarding water usage and potentially puts their health at risk.

- **Inefficient Resource Management:** Without a centralized platform for collecting and analyzing user data, water supply authorities may struggle to gather meaningful insights for efficient resource management and future planning.

1.3 PROJECT OBJECTIVE

- **Improve access to clean water:** The application can be designed to provide information about nearby water sources, including public taps, wells, or water distribution centers. It can help users locate the nearest source of clean water and provide directions to reach there.
- **Real-time water availability updates:** The application can integrate with water supply systems or local authorities to provide real-time updates on the availability of water in different areas. This information can be crucial, especially in areas with limited water resources or during emergencies.
- **Water quality monitoring:** The application can allow users to report water quality issues in their area, such as contamination or discoloration. It can also provide information on water quality standards and guidelines to educate users about potential risks and precautions.
- **Water conservation awareness:** The application can raise awareness about water conservation practices and provide tips on how to reduce water wastage. It can include features like water usage tracking, leak detection, and reminders for efficient water usage.
- **Billing and payment integration:** If the water supply is managed by a utility company, the application can integrate billing and payment features. Users can view their water consumption, receive bills, and make payments conveniently through the mobile app.